Shopee Product & Design Challenge 2022



Shopee Card – New attractive way of contactless payment in Singapore for millennials and Gen Z

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Figma Prototype

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1. Business Problem Definition and Proposed Solution

1.1. Digital Wallet Industry Landscape

Regarding Table 1, there are **huge opportunities for digital wallets** in the world and Asia Pacific. With more than half of world population, Asia Pacific is undoubtedly **the largest** and one of **most promising markets for mobile wallet growth**, boasting many of the most active mobile wallet markets in the world including China and India (Appendix A1). However, **emerging markets** in the region such as Southeast Asia's countries including **Singapore**, Indonesia, Vietnam are potentially the **most important opportunities** for wallet growth and penetration [1].

Region		2020	2025E	CAGR
World	Total users	2.8B	4.8B	▲11.4%
	Txn value	$7.5T	$10T	▲5.9%
Asia Pacific	Total users	1.8B	2.6B	▲7.6%
	Txn value	$4.1T	$7T	▲11.3%

Table 1. Digital wallet industry outlook [1] [2]

Singapore is forecasted **to third fastest growing countries** in the world and **the fastest growing country** in Asia Pacific in **transaction volume of mobile payment** (Appendix A3). Singapore's mobile payments market will **continue to expand** thanks to consumers being heavy users of e-Commerce and **mobile payments** (Table 2). Top players in the market are GrabPay, FavePay, DBS Paylah, and Singtel Dash together accounting for 75% of Singapore digital wallet market (Appendix A2, A4).

Total Users		Penetration	
2020	2025E ▲	2020	2025 ▲
1.8M	5.8M (3.2x)	30.4%	94.7% (3x)
Transaction Volume		**Transaction Value**	
2020	2025E ▲	2020	2025 ▲
101M	1.1B (11x)	1.4B	8B (5.7x)

Table 2. Singapore's estimated digital wallet [1]

1.2. Millennials and Gen Z's Preferences toward Contactless and Cardless Payments

Millennials and Gen Z, generations of creativity and digitalization, will make up 72% of the world's workforce by 2029, of whom it is much important to businesses' long-term success to capture the loyalty [3]. They are **ditching credit cards, physical cards**, and **cash for digital cards and wallets**. Even though they have their physical cards, they **prefer to integrate** card credentials **into Apple/Google wallets** due to security and convenience.

1.3. Competitors & Shopee Analysis

Despite of being emerging and developed market, **Singapore**'s digital wallet industry offers customers with **only few choices of digital cards** including GrabPay and Singtel Dash (Table 3). This is conducive for Shopee to starting its digital card inside ShopeePay wallets.

Based on SWOT Analysis (Appendix A5), Shopee has both strengths and opportunities to excel in digital wallet industry. It is well-known as the no 1 e-commerce platform in Singapore. **Singapore** is among the most developed in the world, and its **people** are among the most **technologically savvy.** It acts as a catalyst **for in-depth penetration of digital wallet and digital card** in long-term projection. ShopeePay is offered a lot of opportunities **to capture emerging and growing market**, which is not consolidated yet with only few competitors.

		GrabPay	fave PAY	dash	DBS PayLah!	Pay
Top up	QR code to top up	✓		✓		✓
	Credit/Debit card top up	✓	✓	✓		✓
	Direct top up by bank transfer	✓	✓	✓	✓	✓
	API connect to bank	✓		✓	✓	
Transfer, payment, Receive and Withdraw	Transfer to bank account	✓	✓	✓	✓	✓
	Transfer to internal user	✓		✓	✓	✓
	Scan to pay	✓	✓	✓	✓	✓
	Receive by QR code	✓		✓	✓	✓
	Withdraw to bank account	✓	✓	✓	✓	✓
Digital card and Financial services	Digital cards	✓		✓		
	Credit-line	✓	✓	✓	✓	✓
	Insurance	✓		✓	✓	
	Investment	✓		✓	✓	

Table 3. Singapore's digital wallet top player analysis

1.4. Proposed Solution and Business Objectives

We proposed that Shopee should **build a new digital card system** inside its ShopeePay system. Our preferred user groups are **millennials and Gen Z in Singapore market**. As Singapore digital wallet market is growing, we believe that **the introduction of Shopee card** would **increase the penetration of ShopeePay** among Singapore's millennials and Gen Z. It allows unbanked customers to have their own cards and benefit from card functionality without the need of registering through traditional banks. The pros and cons of the solutions are analyzed in below table.

Advantages of Shopee cards solution	Disadvantages of Shopee cards solution
(1) Digital cardholders can pay any online or offline 30 million non-Shopee affiliated and Visa-accepted merchants, ShopeePay partner merchants, and e-commerce Shopee.	(1) Customers must be equipped with digital and technology knowledge to utilize all features. Digital cards may not appeal to older demographics.
(2) Digital cardholders can customize their cards based on their preferences of colors and graphics.	(2) ShopeePay is a part of Shopee super-app. ShopeePay has not been offered as a separate and hassle-free application, which is not at the highest level of convenience.
(3) Digital cards offer better security and flexibility for users to create, change their pin, hide their information, lock card instantly, track transactions, and integrate cards with Apple/Google Wallet.	(3) Digital cardholders are susceptible phishing techniques used to gain usernames and passwords. Once a hacker has these details, money can be transferred before users are aware.

2. Product Specifications → Design Mockups → Interactive Prototype

Interactive Prototype: [https://figmashort.link/zpPQyQ]. In case this shorten link doesn't work, please refer to original link in Appendix B.

User Story #1: As a user, I can create my card directly from ShopeePay.

Acceptance Criteria:

- There are explainer pages to educate users about Shopee Card
- System should allow users to verify their identity through Singpass
- Users can set up their PIN and confirm authenticity by mobile OTP
- Users can choose designs and graphics for their cards

User Story #2: As a user, I can view and hide card details (number and CVV).

Acceptance Criteria:

- By clicking on "Card Details", users can see their card number, CVV, validity date on their cards
- By clicking on "Hide Details", users can see their card names

User Story #3: As a user, I can change my PIN in a secure way.

Acceptance Criteria:

- By clicking on "PIN change", users can be directed to confirm their authenticity by current PIN and mobile OTP
- After confirming their authenticity, they can key in and confirm their new PIN by inputting the new OTP twice

User Story #4: As a user, I can lock when losing credentials.

Acceptance Criteria:

- By clicking on "Lock Card", users can be directed to confirm their authenticity by current PIN and mobile OTP in order to lock their cards

User Story #5: As a user, I can unlock when I want to use it again.

- By clicking on "Unlock Card", users can set their PIN and use card as usual after keying in mobile OTP

User Story #6: As a user, I can add my card to Apple Wallet for convenience.

Acceptance Criteria:

- By clicking on "Apple Wallet", users can be directed to Apple Wallet with pre-filled information such as card number, CVV
- Users only need to review their information and click "Confirm" to add their cards to Apply Wallet

User Story #7: As a user, I can report urgent problems when I need.

Acceptance Criteria:

- By clicking on "Report urgent problems", user can view the chatbot Sophie and report problems on this platform

User Story #8: As a user, I can read Q&A about Shopee Pay.

Acceptance Criteria:

- By clicking on "Learn more about ShopeePay", users navigate to help center

User Story #9: As a user, I can see some transactions on card page and see all transactions on another page.

Acceptance Criteria:

- On the landing page of card, users can see 4 latest transactions
- By clicking on "See all", users can see all transactions they executed

User Story #10: As a user, I can change my card designs to my preferences.

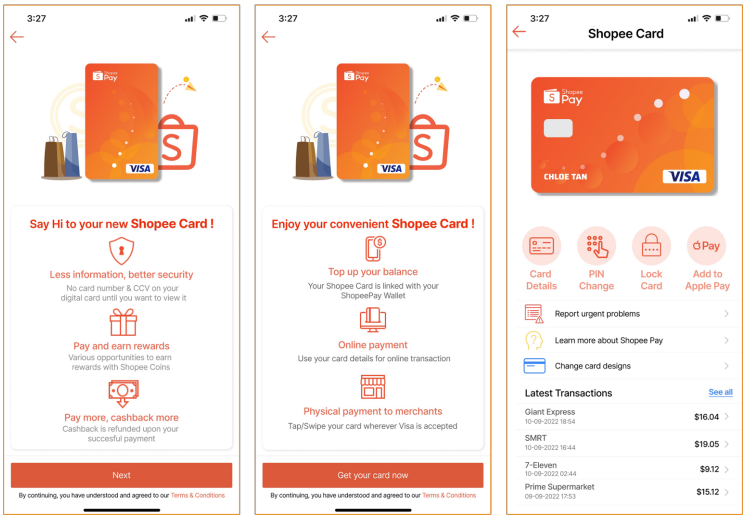
Acceptance Criteria:

- By clicking on "Change card design", users can change color and graphic of their cards. They reselect other designs when they change their minds
- Customers can change their selection without the need of deselecting

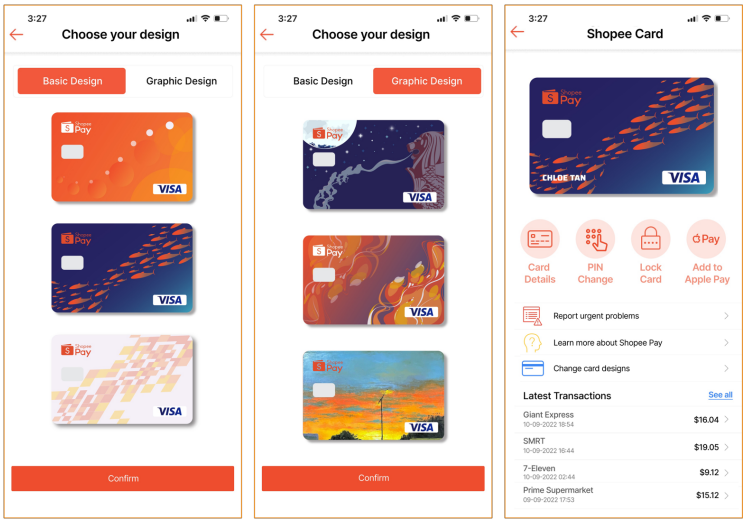
Design Mockups of Main Functionality

For user journey of individual User Story 1 – 9, please refer to appendix B.

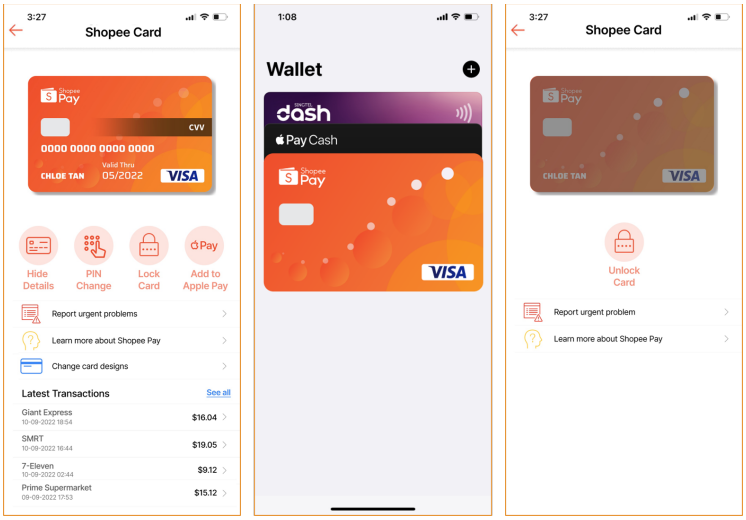
Following design mockups are main designs of product's functionality



Educational explainer pages and main pages of Shopee Card



Shopee allows users to change card design in a convenient way



Users integrate to Apple Pay, lock/unlock cards, change PIN, and more...

3. Business Impacts on ShopeePay's business

ShopeePay Card is expected to exert substantial effects on ShopeePay's business and operations.

(1) **Revenue stream**: It creates **a new revenue stream from Visa-accepted merchants** where ShopeePay QR's code is not accepted yet.
(2) **Awareness and popularity**: It allows ShopeePay to **go beyond Shopee's e-commerce ecosystem**. Currently, ShopeePay is mainly used for eCommerce and ShopeePay-accepted merchants. When the card is introduced, digital cards are used in **(1)** commuting on bus and MRT, **(2)** shopping and dining at non-ShopeePay partner merchants, **(3)** international payments both online and offline. ShopeePay can **gain citizen's awareness and recognition** that it is not just an eCommerce wallet but **a powerful digital payment vendor**.
(3) **Strong and flexible cashflows**: The more frequently Shopee cards are used, **a stronger cash flows** ShopeePay has in account. This allows ShopeePay to **mobilize money for other operations** and **investments** on other business lines.

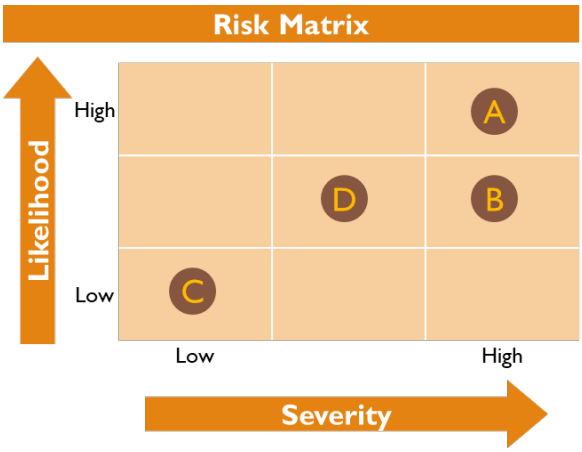
4. Performance Evaluation Metrics

Successful metrics are important to **design product's tracking tools** inside product and **evaluate the performance of products**. For ShopeePay's cards, following metrics are used to assess product' s performance and for future improvements.

(1) Percentage of Market Share
(2) Daily/Monthly Active Users
(3) Average Daily/Monthly Number of Transactions per User
(4) Average Daily/Monthly Transaction Value
(5) Average Session Length – Measure average amount of time spent in app per user
(6) Customer Lifetime Value – Total user spend over total number of buyers to date
(7) Customer Retention Rate
(8) Customer Journey Analysis – Analyze customer journey to detect pain points and improve experiences
(9) Customized Cards Usage - % of customers use each type of cards

5. Risks of ShopeePay Card Solution

Risks	Mitigation
(A) Economic risks: Economic downturns slow consumers' expenses and the economy.	Do market research carefully to assess the growth and future of digital wallet market.
(B) Compliance risk: Tight KYC and compliance requirements from regulations pose a significant risk.	Develop a thorough KYC procedures ensure that it is in line with compliance and regulations.
(C) Operation risk: It is challenging to capture market shares of top 3 big players.	Since market is growing, it is a must for an excellent card service to capture market shares.
(D) Security and Data: Personal information and financial data are vulnerable to data leak & breaching.	Develop comprehensive layers of verification for accessing data such as OTP, Authenticator.



6. Catalysts of ShopeePay Card Solution

There are some catalysts for ShopeePay Card's product launch and success.

(1) **Higher-than-estimated growth rate** of Singapore's digital wallet industry paves the way for adoption and penetration of ShopeePay among millennials and gen Z.
(2) ShopeePay card **User Interface and User Experience (UI&UX) are better** compared with competitors. This is essential in **customer acquisition**, especially younger generations who are into user-friendly applications.
(3) **Value-added services** such as **loyalty rewards and credit**, non-traditional services not offered by banks, are attractive to **capitalize on a highly active user base**. It would lead to higher revenue and profitability.
(4) Successful of ShopeePay Card can **pave the way for other financial service products** such as credit line on cards, buy now pay later, and cryptocurrencies. Those products are essential to profitability and become a competitive player.
(5) **APIs** that connect financial institutions to local retailers will be **key to the international and local growth of digital wallet usage** for large, cross-border eCommerce merchants. Better APIs technology, better growth and expansion.

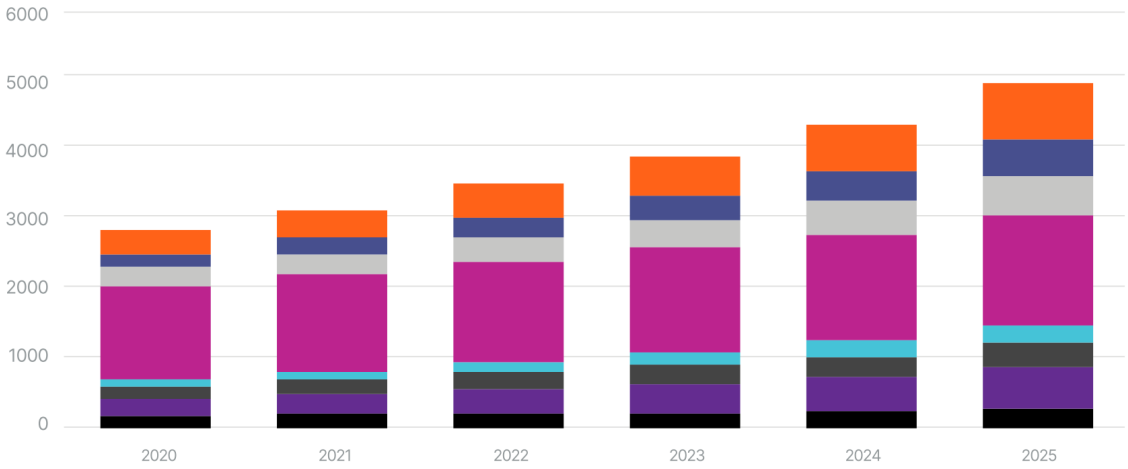
7. Business Objective Evaluation

The objective of **boosting the penetration rate of Shopee Pay** is likely to be **achieved** since (1) Singapore's digital wallet industry is experiencing **strong growth** during next 5 years (2) Industry is **not consolidated** yet since the industry is at early state and has a few main players (3) Digital card is **innovative** and in line with ShopeePay's direction for its future digital wallet (4) Target audience, Singapore's millennials and Gen Z, have strong **preferences** toward digital wallet and cardless payments.

Appendix

Appendix A: Research and Data

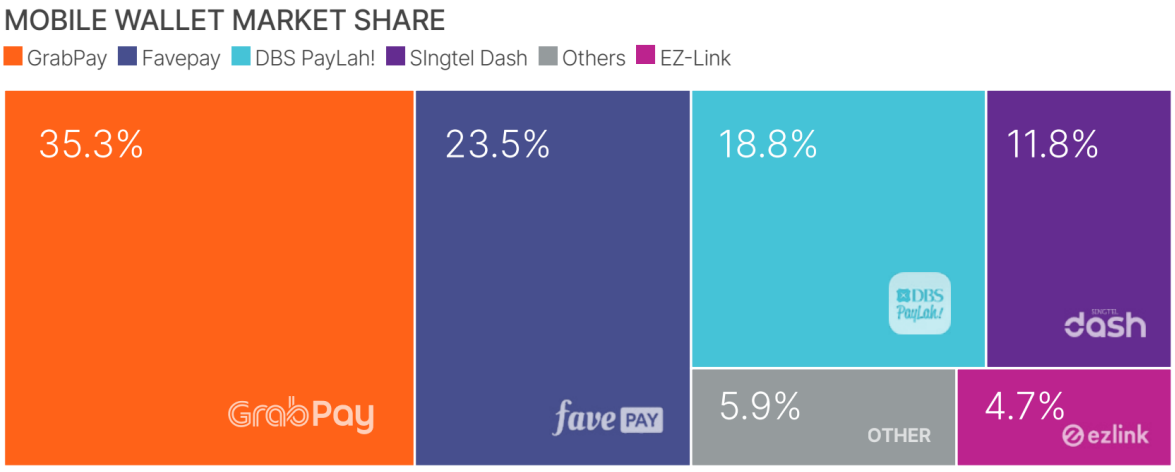
A.1. Majority of World's Total Mobile Wallet in Use is Asia Pacific



	2020	2021	2022	2023	2024	2025
North America	184.7	201.0	219.9	233.2	247.6	275.4
Latin America	227.3	277.5	336.6	406.9	494.9	605.7
West Europe	200.1	216.3	245.1	257.0	285.3	331.9
Central & East Europe	76.3	103.2	143.0	187.5	215.0	248.9
Far East & China	1,343.4	1,386.9	1,429.8	1,466.5	1,501.8	1,541.4
Indian Subcontinent	269.2	287.4	333.2	393.7	465.6	550.4
Rest of Asia Pacific	179.7	223.1	281.7	349.9	426.5	520.7
Africa & Middle East	322.9	390.0	471.4	561.5	669.6	798.2
Global	**2,803.7**	**3,085.5**	**3,460.6**	**3,856.5**	**4,306.3**	**4,872.7**

Sources: Mobile Wallet Report [1]

A2 – Singapore's Mobile Wallet Market Share



Sources: Mobile Wallet Report [1]

A3 – Fastest Growing Countries for Mobile Payment – Transaction Volume

Over the next five years, mobile payment transaction volume (in millions of $) by country is projected with growth rates.

		2020	2025	CAGR
1	Russia	$9,166.7	$65,101.2	48.0%
2	Brazil	$19,844.9	$130,708.3	45.8%
3	Singapore	$1,161.2	$7,565.7	45.5%
4	Germany	$17,120.4	$105,219.8	43.8%
5	Malaysia	$4,584.2	$25,946.0	41.4%
6	Portugal	$1,888.5	$9,823.4	39.1%
7	Philippines	$13,145.3	$59,312.4	35.2%
8	Sweden	$6,818.9	$29,978.4	34.5%
9	Mexico	$19,040.6	$83,687.5	34.5%
10	Indonesia	$22,968.9	$98,093.6	33.7%
11	France	$16,822.7	$66,892.6	31.8%
12	Norway	$2,876.7	$10,825.6	30.4%
13	Thailand	$8,827.3	$33,104.6	30.3%
14	Vietnam	$11,906.8	$44,049.2	29.9%
15	Spain	$13,319.6	$40,110.7	24.7%
16	Denmark	$4,273.4	$12,483.0	23.9%
17	Saudi Arabia	$8,332.2	$23,449.4	23.0%
18	Netherlands	$7,317.6	$20,371.0	22.7%
19	UK	$48,054.7	$116,927.4	19.5%
20	India	$167,579.8	$401,690.4	19.1%
21	Australia	$23,250.9	$54,324.5	18.5%
22	Canada	$30,630.6	$68,994.6	17.6%
23	Pakistan	$30,480.9	$66,262.8	16.8%
24	US	$469,187.1	$1,018,392.2	16.8%
25	Japan	$166,019.1	$310,889.9	13.4%
26	United Arab Emirates	$2,247.3	$4,018.7	12.3%
27	Taiwan	$28,709.3	$47,835.9	10.8%
28	South Korea	$66,246.7	$108,549.5	10.4%
29	Egypt	$21,254.9	$34,826.6	10.4%
30	Nigeria	$44,335.4	$71,003.0	9.9%
31	China	$3,377,408.9	$5,344,518.5	9.6%
32	Kenya	$110,987.5	$160,053.4	7.6%

Sources: Mobile Wallet Report [1]

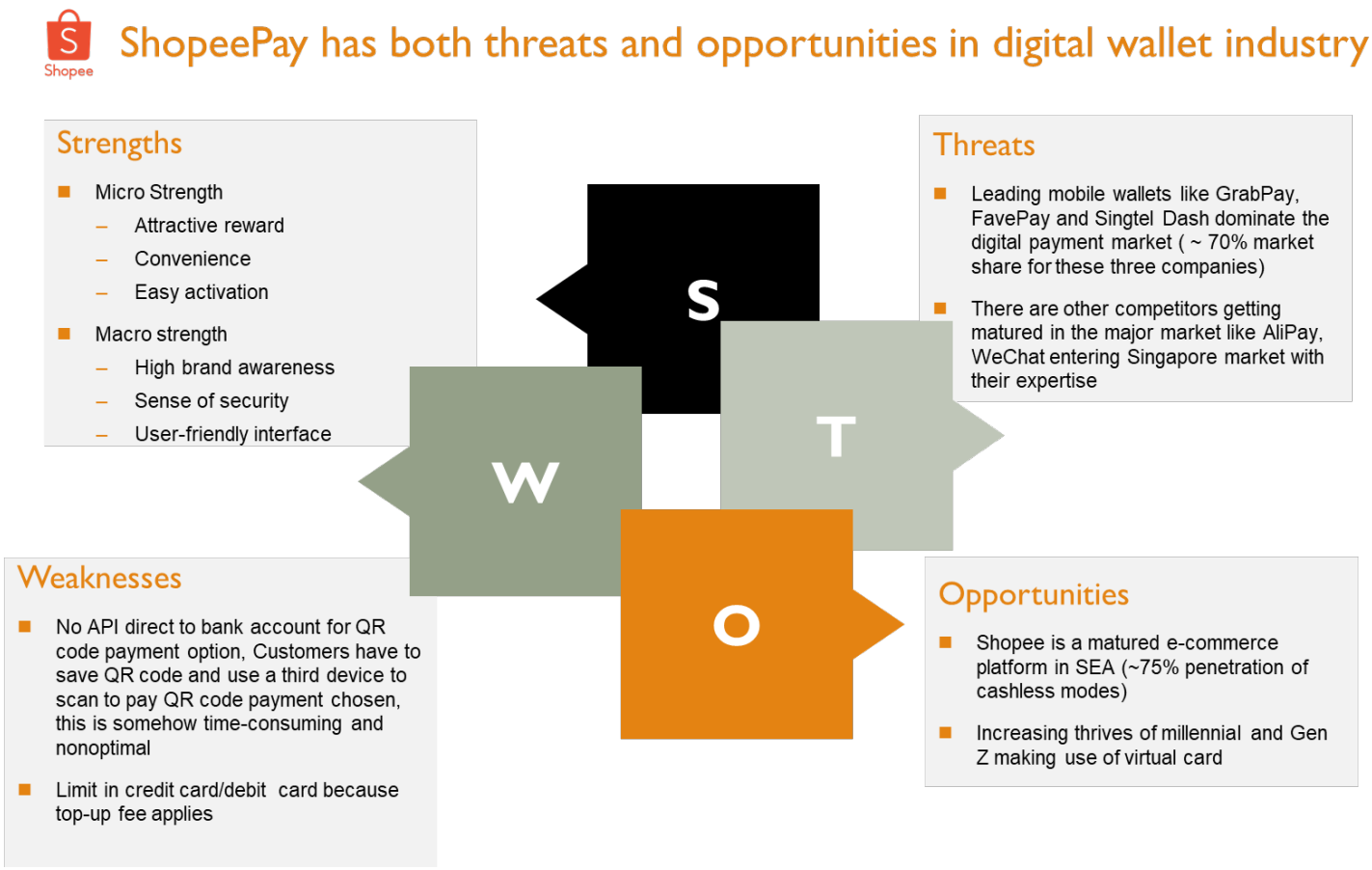
A4 – Fastest Growing Mobile Wallets

Over the next five years, transactions processed by mobile wallet (in millions of $) is projected with growth rates.

			2020	2025	CAGR
1	SadaPay	Pakistan	$65.6	$6,066.1	147.3%
2	MercadoPago	Brazil	$4,073.7	$43,156.6	60.3%
3	PicPay	Brazil	$9,877.2	$59,123.6	43.0%
4	GrabPay	Singapore	$480.8	$2,838.8	42.6%
5	Favepay	Singapore	$320.5	$1,892.5	42.6%
6	DBS PayLah!	Singapore	$256.4	$1,514.0	42.6%
7	YooMoney	Russia	$3,964.3	$22,662.3	41.7%
8	QIWI	Russia	$3,267.3	$18,677.7	41.7%
9	PayPal	Russia	$435.6	$2,490.4	41.7%
10	WebMoney	Russia	$5,086.7	$29,078.1	41.7%
11	Boost	Malaysia	$1,240.7	$6,773.8	40.4%
12	GrabPay	Malaysia	$2,114.9	$10,391.6	37.5%
13	Touch N Go	Malaysia	$2,002.1	$9,687.3	37.1%
14	GrabPay	Philippines	$1,136.5	$5,400.1	36.6%
15	PayMaya	Philippines	$6,364.6	$26,644.6	33.2%
16	MercadoPago	Mexico	$7,546.9	$31,307.0	32.9%
17	RappiPay	Mexico	$4,025.0	$16,697.1	32.9%
18	PayPal	Mexico	$2,264.1	$9,392.1	32.9%
19	G-Cash	Philippines	$7,501.1	$30,557.1	32.4%
20	Klip	United Arab Emirates	$418.5	$1,595.2	30.7%
21	PayPal	Brazil	$1,279.6	$4,869.0	30.6%
22	OVO	Indonesia	$10,753.6	$40,907.7	30.6%
23	Shopeepay	Indonesia	$4,399.2	$16,735.0	30.6%
24	LinkAja	Indonesia	$3,910.4	$14,875.5	30.6%
25	Gojek	Indonesia	$3,714.9	$14,131.8	30.6%
26	DANA	Indonesia	$3,421.6	$13,016.1	30.6%
27	Truemoney Wallet	Thailand	$5,591.1	$19,298.3	28.1%
28	LINE Pay	Thailand	$2,625.1	$9,060.8	28.1%
29	GrabPay	Thailand	$430.1	$1,484.5	28.1%
30	AirPay	Thailand	$602.1	$2,078.3	28.1%
31	mPay	Thailand	$516.1	$1,781.4	28.1%
32	Momo	Vietnam	$7,503.1	$25,765.0	28.0%

Sources: Mobile Wallet Report [1]

A5 – SWOT Analysis of ShopeePay

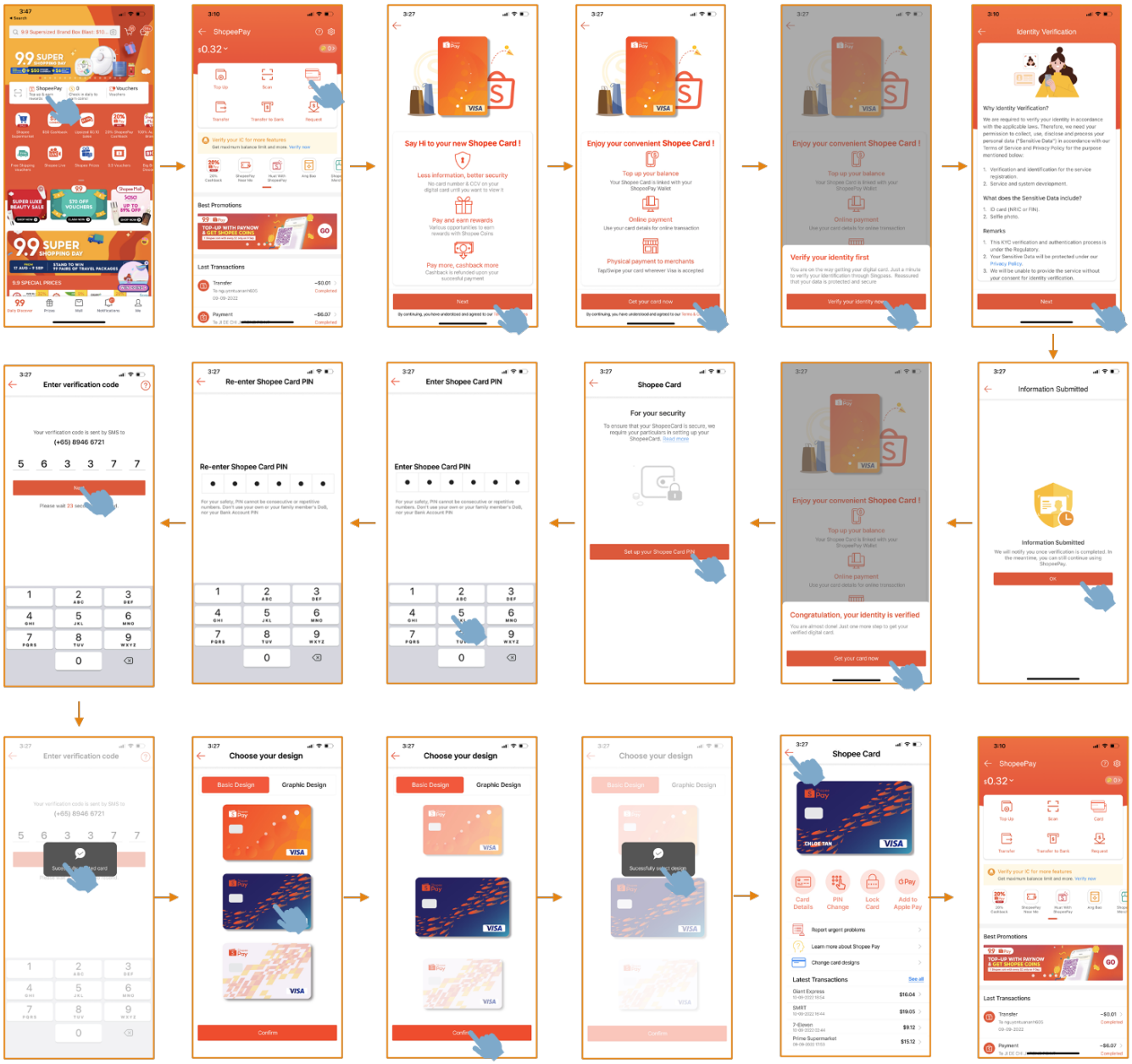


Sources: Team's analysis

Appendix B: User Story and Customer Journey

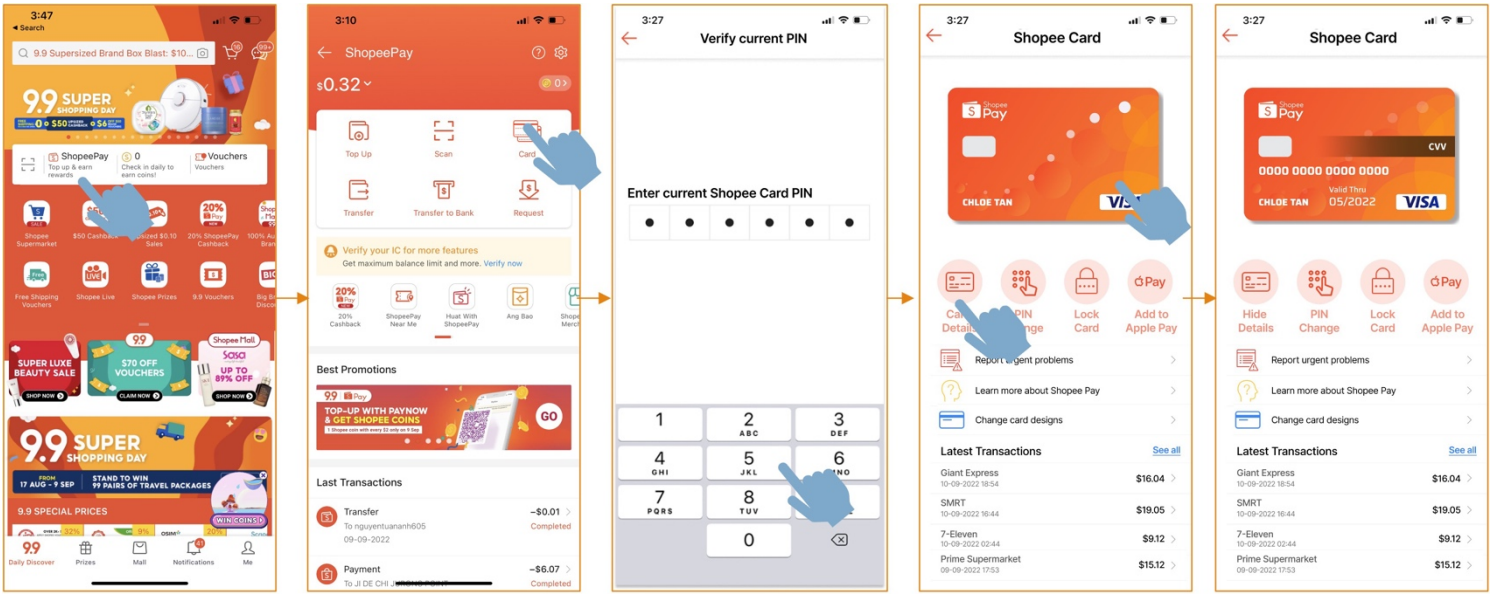
Original link of Figma: https://www.figma.com/proto/ZAebqMLvFDRzr3h31G9hzh/%5BTeam-4222%5D-Prototype?node-id=557%3A2&starting-point-node-id=557%3A2 Please access this link in case that other methods doesn't work.

User Story 1: As a user, I can create my card directly from ShopeePay



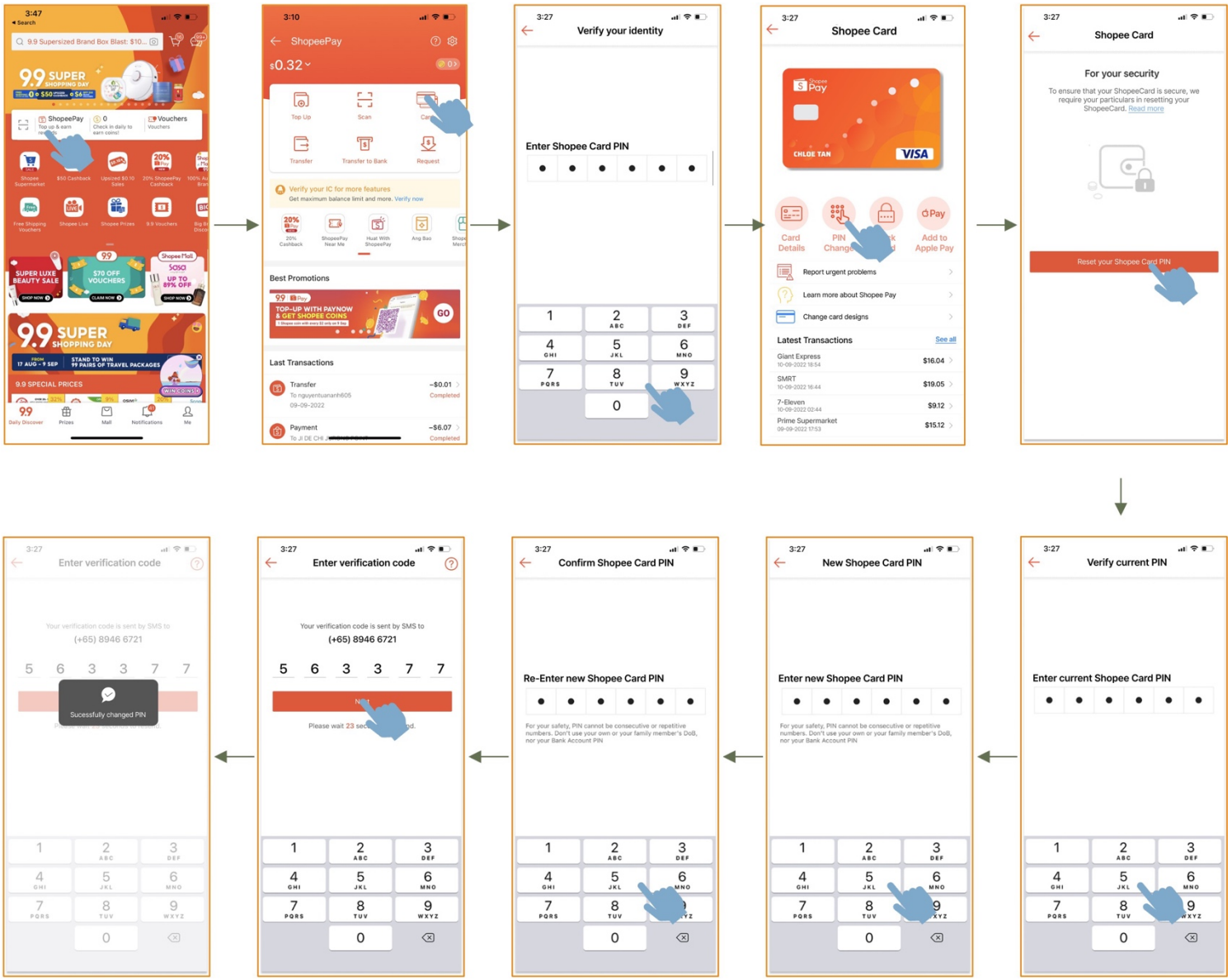
1. When users create cards, they will be navigated to explainer pages to educate users about Shopee Card's feature.
2. Customers have to verify their identity by Singpass before they are able to create cards.
3. Users can set their PIN by inputting PIN, confirming PIN, and entering mobile OTP.
4. Users can select card design from a list of basic design and graphic design.
5. (Optional) Users can select other designs without the need of deselecting old selection.
6. Users can exit from card landing page by using top left arrow.

User Story 2: As a user, I can view and hide card details: number and CVV.



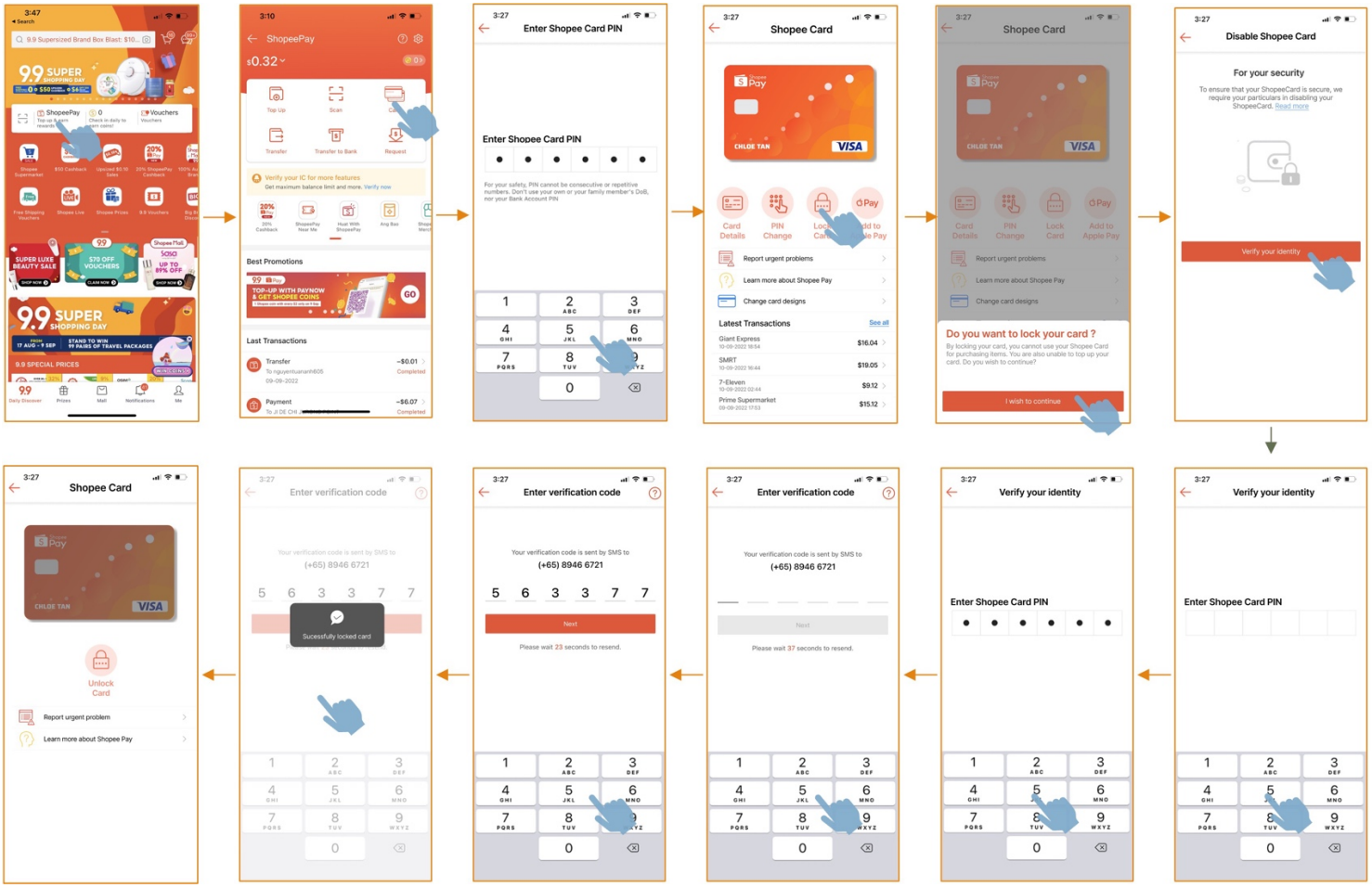
1. When entering card landing page, users are required to enter PIN for security purpose.
2. Subsequently, they can see their card details by clicking on “Card details” or the card itself.
3. Additionally, users can hide details by clicking on “Hide Details” or the card itself.

User story 3: As a user, I can change my PIN in secure way.



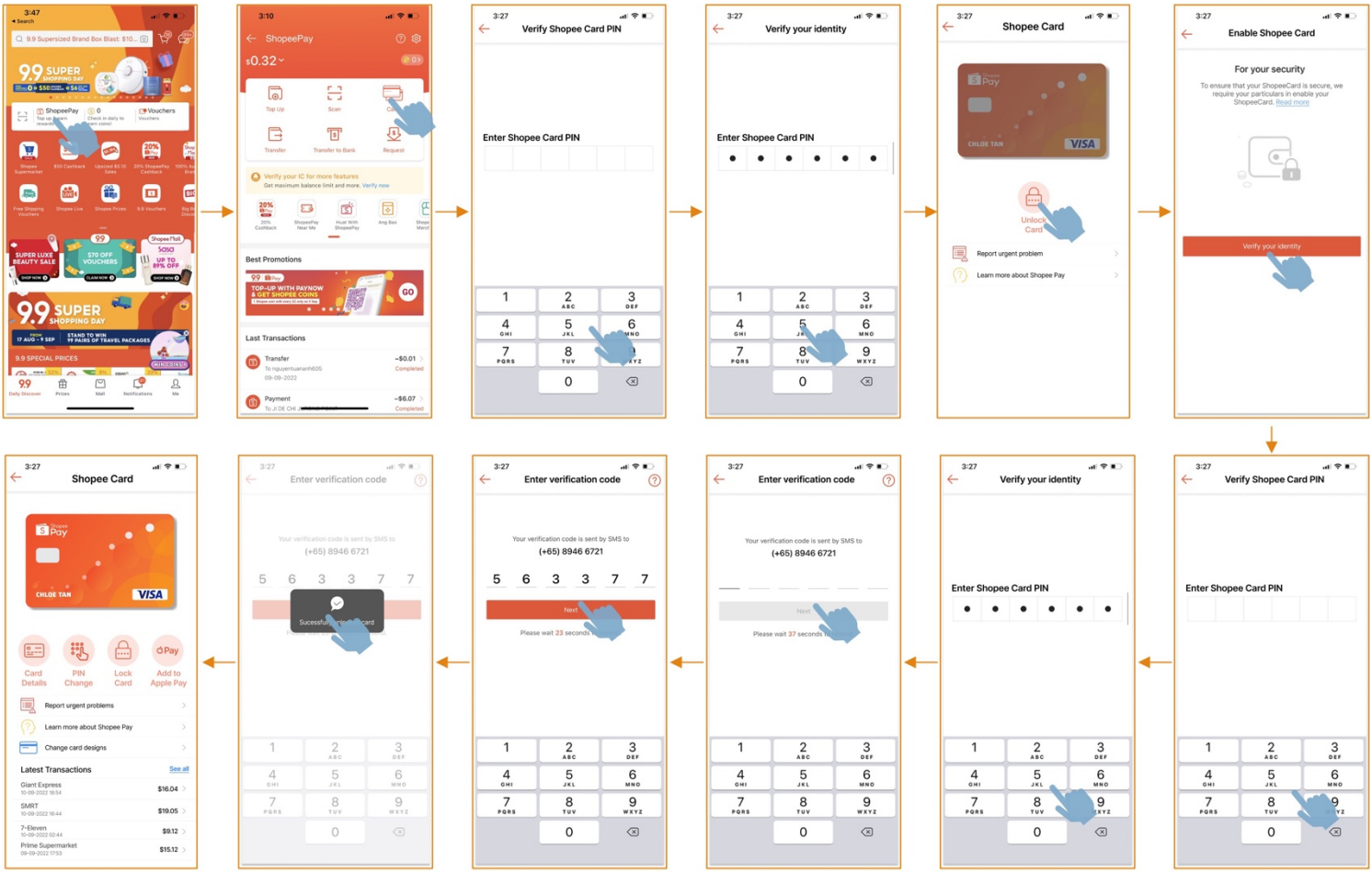
1. When entering card landing page, users are required to enter PIN for security purposes.
2. Users can change their PIN by selecting “PIN change”
3. Users are required to enter PIN 2 times → new PIN 2 times → mobile OTP to complete PIN change process.

User story 4: As a user, I can lock my card when losing credentials.



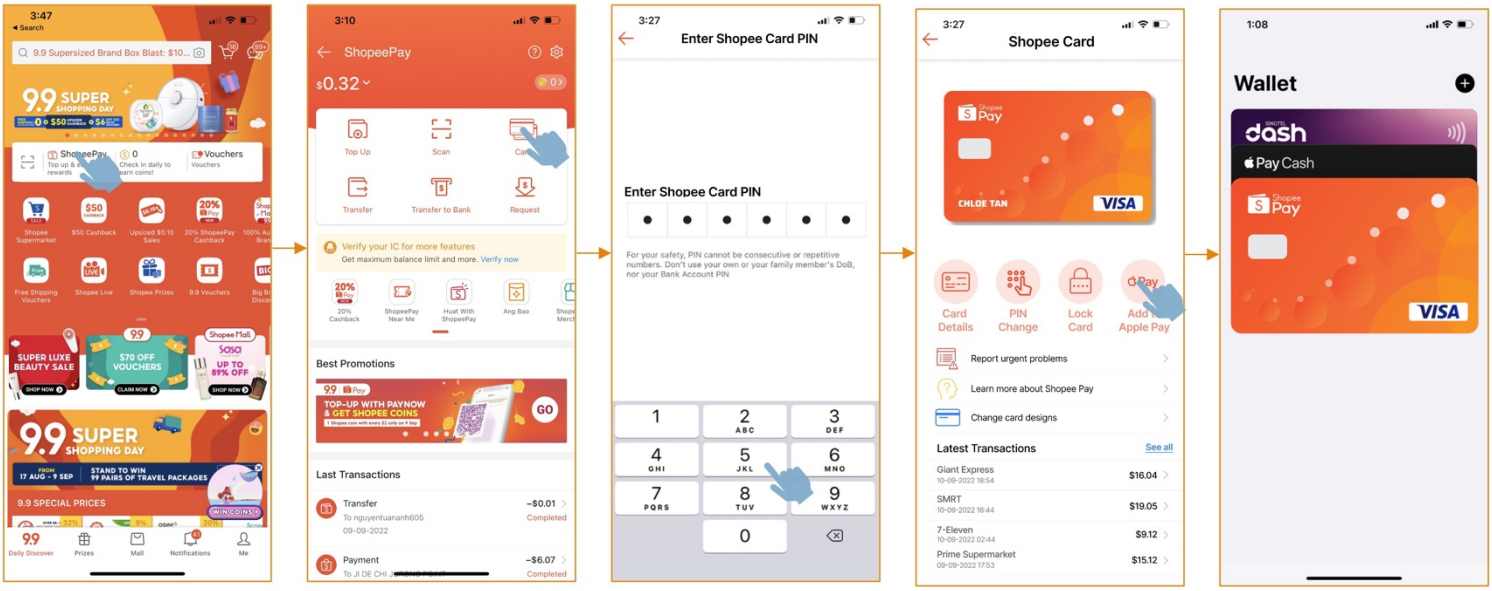
1. When entering card landing page, users are required to enter PIN for security purposes.
2. Users can lock their cards by clicking on “Lock Card”.
3. Users are required to enter PIN → mobile OTP to complete lock card process.

User story 5: As a user, I can unlock my card when I want to use it again.



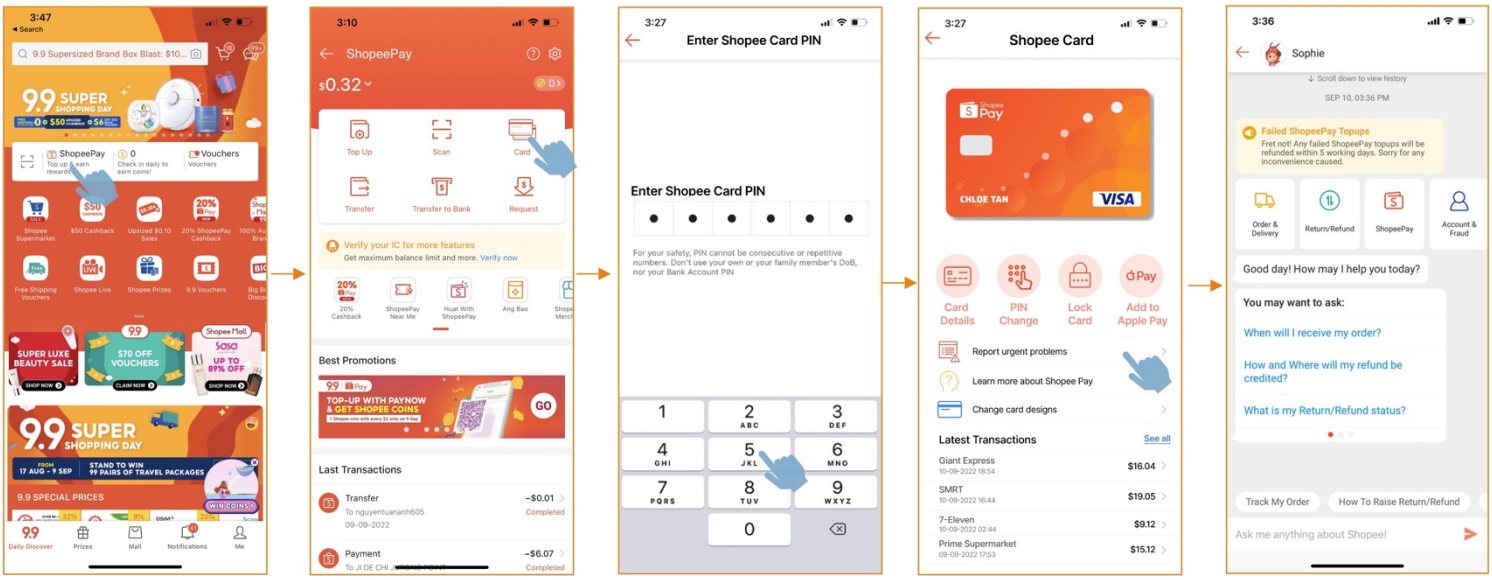
1. When entering card landing page, users are required to enter PIN for security purposes.
2. Users can unlock their cards by clicking on “Unlock Card”.
3. Users are required to enter PIN → mobile OTP to complete unlock card process.

User story 6: As a user, I can add my card to Apple Wallet for convenient use.



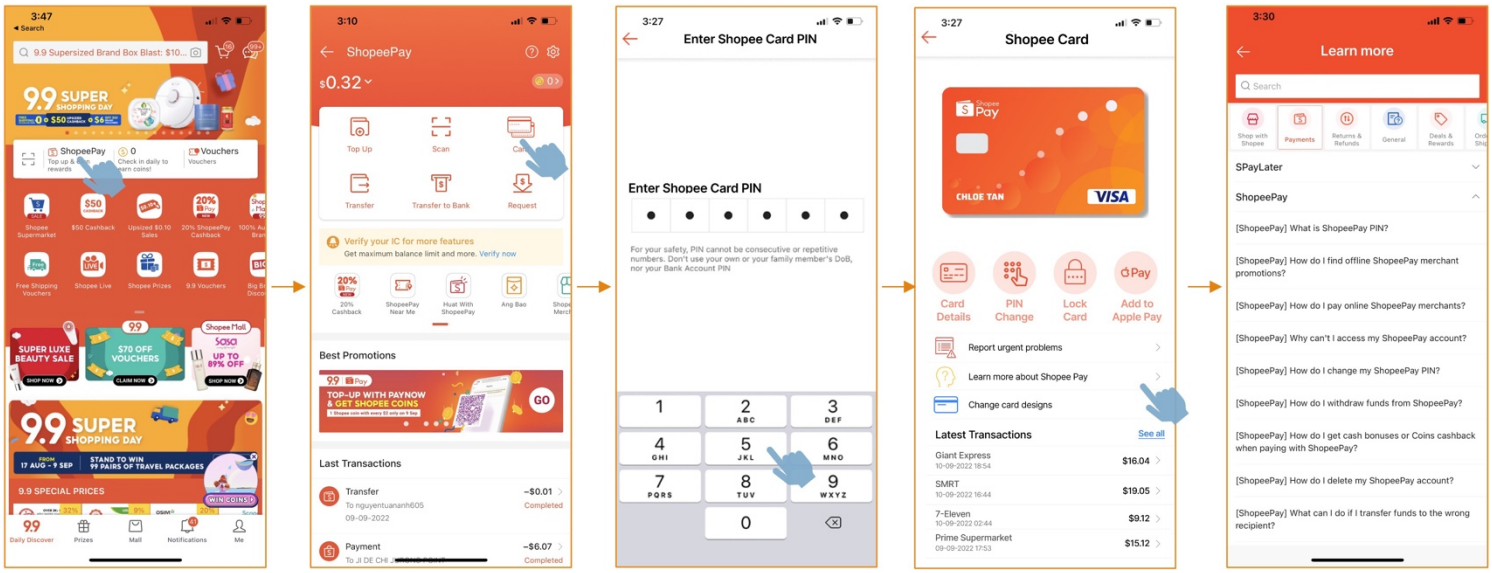
1. When entering card landing page, users are required to enter PIN for security purposes.
2. Users can add cards to Apple Pay by clicking on “Add to Apple Pay”
3. Users are navigated to Apple Pay dashboard.

User story 7: As a user, I can report urgent problem when I need.



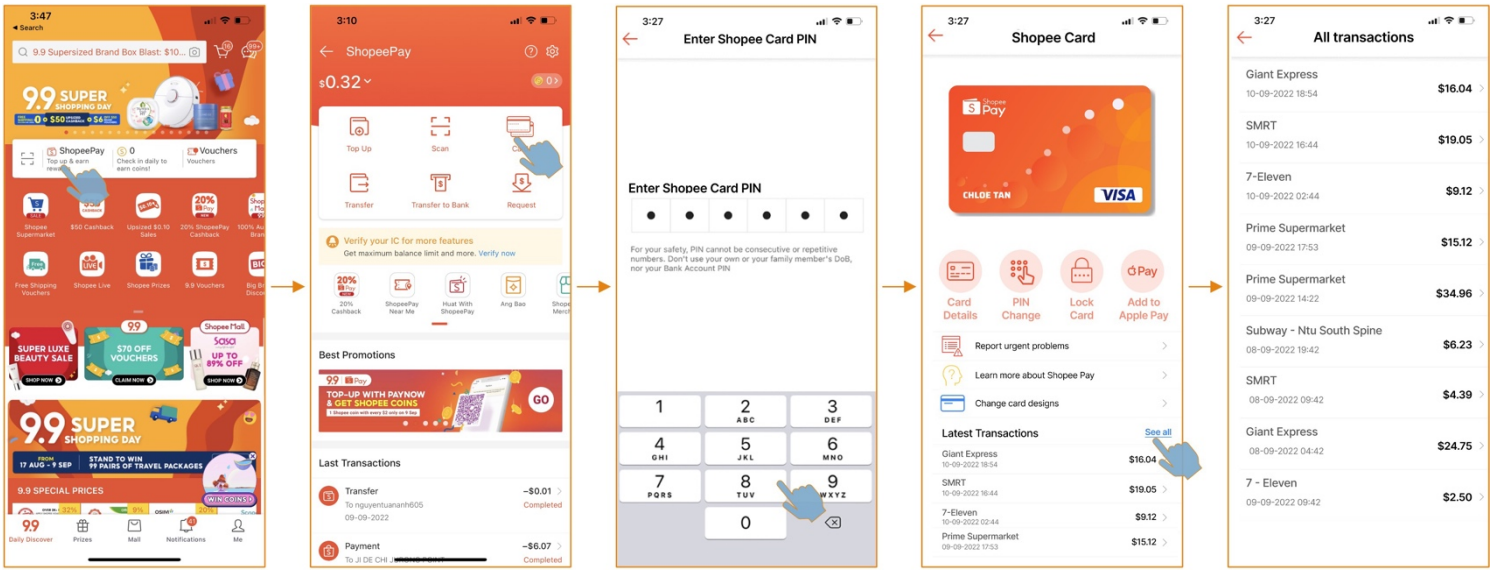
1. When entering card landing page, users are required to enter PIN for security purposes.
2. Users can report their problems by clicking on “Report urgent problems”
3. Users are navigated to help center with virtual assistant Sophie.

User story 8: As a user, I can learn more about ShopeePay.



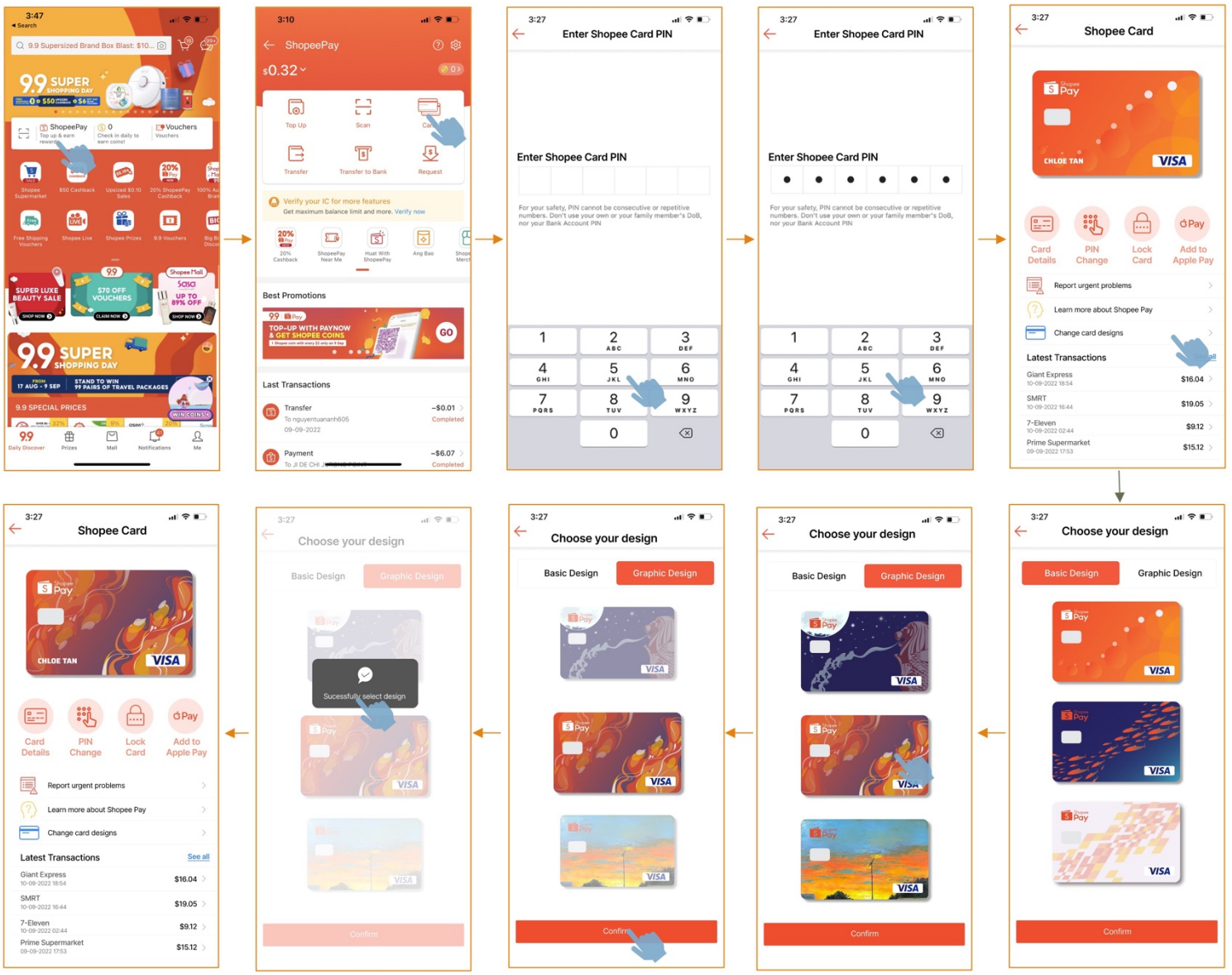
1. When entering card landing page, users are required to enter PIN for security purposes.
2. Users can read more information about ShopeePay by clicking on “Learn more about Shopee Pay”
3. Users are navigated to Q&A center to see all questions’ answers.

User story 9: As a user, I can see lastest transactions on card page and see all transactions on another page.



1. When entering card landing page, users are required to enter PIN for security purposes.
2. Users can see all transactions on cards by clicking on “See all”
3. Users are navigated to all transactions dashboards to see all transactions relating to cards.

User story 10: As a user, I can change my card design.



1. When entering card landing page, users are required to enter PIN for security purposes.
2. Users can navigate to change card design by clicking on “Change card designs”.
3. Users can select cards on basic and graphic designs.
4. Users can select other card designs without the need of deselecting previous choices.

References

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